

1. Explain the purpose and importance of the information gathering phase in security testing.
2. Describe the difference between passive and active information gathering.
3. Explain the main types of information that are targeted during reconnaissance.
4. Describe the role of DNS in Internet architecture and in infrastructure discovery.
5. Explain DNSenum, DNSrecon, and DNSmap and their functions.
6. Describe the concept of subdomains and their impact on system exposure.
7. Explain what a DNS zone transfer is and why it has security relevance.
8. Explain the role of TTL and caching in DNS operation.
9. Describe the concept of load balancing and its primary purpose.
10. Explain what a CDN is and why it is used in modern web architectures.
11. Describe the function of a reverse proxy in protecting and managing web traffic.
12. Explain what a WAF is and its role.
13. Describe the concepts of IDS and IPS and the fundamental difference between them.
14. Explain the importance of network segmentation in network security.
15. Explain what live host discovery is and why it is performed.
16. Describe the importance of traffic control at the network level.
17. Explain what vulnerability analysis is and its purpose.
18. Describe the difference between vulnerability scanning and exploitation.
19. Explain the concept of fuzzing and why it is used in application security.
20. Describe what privilege escalation is and why it is critical.
21. Explain why misconfigurations represent a security risk.
22. Describe the concept of directory enumeration.
23. Explain the most common vulnerabilities in web applications.
24. Describe the importance of security headers in web applications.
25. Explain the risk of information exposure through error messages.
26. Describe the concept of brute-force and dictionary attacks.
27. Explain the importance of password hashing and password policies.
28. Describe the role of rate limiting in protecting services.
29. Explain why security must be part of system design.
30. Explain how modern web architectures affect the real external visibility of a system.
31. Explain how the combination of CDN, WAF, and load balancing changes the approach to security assessment.
32. Explain why the lack of information during reconnaissance is often an indicator of good security.
33. Explain the limits of information gathering in fortified environments.
34. Explain the trade-off between availability and security in public systems.
35. Explain why perimeter-only security no longer works.
36. Explain the importance of segmentation in limiting attack propagation.
37. Explain why monitoring is as important as prevention.
38. Explain why configuration-related vulnerabilities are difficult to manage.
39. Explain the role of fuzzing in discovering undocumented vulnerabilities.
40. Explain why privilege escalation represents a systemic failure.

41. Explain the importance of the principle of least privilege.
42. Explain the impact of missing logging on incident investigation.
43. Explain why complex architectures require integrated security strategies.
44. Explain the risk of indirect information exposure.
45. Explain why security should be viewed as an architectural problem.
46. Explain the role of organizational policies in technical security.
47. Explain why technology alone is insufficient without clear processes.
48. Explain the relationship between system design and the attack surface.
49. Explain why the automation of attacks changes the security landscape.
50. Explain the importance of defense in depth.
51. Explain why reactive security is always insufficient.
52. Explain why security testing must be cyclical.
53. Explain the impact of human factors on security architecture.
54. Explain why technical decisions have long-term security consequences.
55. Explain the risk of dependency on external technologies.
56. Explain why the exposure of auxiliary services is critical.
57. Explain why security is an interdisciplinary responsibility.
58. Explain why small failures often lead to major incidents.
59. Explain why security should be treated as a strategic process.
60. A command was executed: `fping -a -g 192.168.1.0/24`.
What process is performed, and what information does this command provide?
61. Ping was used with the `-c` parameter against a specific host.
What is being tested, and what does the result indicate?
62. TCP SYN traffic was sent to port 80 using `hping3`.
What type of test was performed, and why is this method used?
63. ARPing was used against an IP address in the local network.
What information is intended to be obtained, and why does this command work only in a LAN?
64. Nmap was executed with the `-sn` option over a subnet.
Which testing phase does this represent and what result is expected?
65. A TCP SYN scan was performed with `nmap` on specific ports.
What information does this scan provide?
66. Dig was used to perform a DNS query for a domain.
What information does this command return?
67. A dig query was performed for MX records of a domain.
What is being discovered, and why is it important from a security perspective?
68. A DNS zone transfer was attempted using `dig axfr`.
What is being attempted, and why is this related to information leakage?
69. Whois was used for a public domain.
What type of information does this command provide during information gathering?
70. The LBD tool was executed against a domain.
What is being analyzed, and what does this test aim to identify?
71. Wafw00f was used against a website.
What type of information does this tool attempt to identify?

72. Wafw00f was used with the -a parameter.
What is the difference compared to standard usage?
73. Dirb was executed against a URL.
What process is performed and what risk is associated with the results?
74. Ffuf was used with a wordlist for URL fuzzing.
What is being tested, and why are wordlists used?
75. A Nikto scan was performed against a web server.
What types of vulnerabilities does this tool aim to identify?
76. WPScan was used against a website.
What type of system is targeted and what information is sought?
77. Tcpdump was executed to monitor network traffic.
What type of information can be analyzed using this tool?
78. Wireshark was used for traffic analysis.
What types of analysis are performed, and why is this tool important?
79. SPIKE Generic was used to send traffic to a service.
What type of test is performed, and what does it aim to discover?
80. Unix-privesc-checker was executed in standard mode.
What does this script analyze, and in which phase is it used?
81. Files with SUID permissions were searched on the system.
Why is this analysis related to privilege escalation?
82. John the Ripper was used with a wordlist.
What process is performed, and what type of attack does this represent?
83. Hashcat was used to crack hashes using a dictionary attack.
What does this indicate about password storage practices?
84. Hydra was used to test SSH credentials.
What is being attempted, and why is this considered an online attack?
85. Medusa was used to test passwords against an FTP service.
What risk does this command demonstrate?
86. SEToolkit was executed in Kali Linux.
What types of attacks does it simulate, and why is it included in security assessments?

Note : Questions requiring you to explain the command, the output in Kali Linux, and why the command was used will be based on the examples covered in the lecture.